

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/230964484>

The use of the Allan deviation for the measurement of the noise and drift performance of microwave radiometers

Article in *Measurement Science and Technology* · May 2007

DOI: 10.1088/0957-0233/18/7/018

CITATIONS

39

READS

3,020

3 authors:



David V Land

University of Glasgow

37 PUBLICATIONS 651 CITATIONS

[SEE PROFILE](#)



A. P. Levick

Freelance science teacher

31 PUBLICATIONS 465 CITATIONS

[SEE PROFILE](#)



Jeff Hand

King's College London

183 PUBLICATIONS 4,352 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Inverse Problems and Parameter Estimation in Engineering [View project](#)



RF safety of neonates at 7T [View project](#)

The use of the Allan deviation for the measurement of the noise and drift performance of microwave radiometers

D V Land¹, A P Levick² and J W Hand³

¹ Department of Physics and Astronomy, University of Glasgow, Glasgow G12 8QQ, UK

² Thermal Metrology, National Physical Laboratory, Teddington TW11 0LW, UK

³ Division of Clinical Sciences, Imperial College, Hammersmith Hospital, Du Cane Road, London W12 0NN, UK

E-mail: d.land@physics.gla.ac.uk, Andrew.levick@npl.co.uk and j.hand@imperial.ac.uk

Received 29 January 2007, in final form 23 March 2007

Published 15 May 2007

Online at stacks.iop.org/MST/18/1917

Abstract

The use of the Allan deviation for the analysis of signal noise and drift components is considered in the context of microwave radiometry. The noise behaviour of two types of microwave radiometer is modelled and compared with measurements of the performance of these radiometers analysed using the Allan deviation method.

Keywords: Allan deviation, microwave radiometry, noise signal analysis

1. Introduction

All measurement systems have a measurement resolution that ultimately must be limited by thermally induced random fluctuations, ‘noise’, of the measured quantity, and practical systems will also experience some degree of variation with time of parameters which affect the value of the measured quantity, ‘drift’ and limit measurement accuracy. Both noise and drift can take a variety of forms having different measurement time or frequency dependences. Gaussian or ‘white’ noise, flicker or ‘ $1/f$ ’ noise and random-walk drift are examples commonly met in electronic measurement devices. To understand system performance the noise and drift of a measurement quantity must be analysed in a way that allows identification of causal sources and determination of their magnitude. This is usually assisted by determination of the time or frequency dependence of the noise and drift components present in the value of the measured quantity. The commonly used standard deviation measure of variation does not provide a simple way to distinguish noise or drift types, with the magnitudes of these components difficult to assess when they are overlaid in a spectral power density plot. In contrast, the Allan deviation provides, directly, magnitude versus time separation which in the form of a log–log deviation data plot allows the different noise and drift types to be readily identified by the slopes of the different plot regions (Allan 1966, 1987, Levine 1999).

The Allan deviation method has been very extensively applied to the measurement of atomic clock stability but its application to noise signal amplitude analysis has been very limited (Allan 1987, Huntley 1988, Park *et al* 1991, Goodberlet and Mead 2006). It is here applied to the analysis of the relatively complex noise behaviour of the temperature signal from two types of microwave radiometer designed to suit the rather difficult measurement requirements of several medical and industrial applications (Hand *et al* 2001, Land 2001).

Microwave radiometric temperature measurement is a technique where Gaussian thermal noise inherent in the measurement and the presence of instrument drift due to environmental temperature changes impose significant practical limitations on measurement resolution and accuracy. This is particularly the case for medical and industrial applications where microwave radiometry is used to provide non-invasive temperature estimates within tissues and other materials and for which the minimum possible measurement times must be used (Carr *et al* 1981, Chive *et al* 1984, Leroy *et al* 1987, 1998, Land 1987, Foster and Cheever 1992). The limitations imposed by noise and drift are seen particularly acutely in multi-frequency radiometry used to estimate internal temperature profiles in materials. The accuracy of the temperature estimation possible is here directly limited by the radiometer measurement performance of each of the several measurement channels (Mizushina *et al* 1989, Maruyama *et al* 2000, Hand *et al* 2001, Bardati

et al 2004, Sugiura *et al* 2004). The Allan deviation analysis technique provides quantitative measurement of the required performance information in a form appropriate to this type of instrumentation and applications.

2. Sources of microwave radiometer noise and drift

2.1. Inherent radiometric signal noise

All radiometers make a measurement of a randomly fluctuating wideband noise signal, imposing some degree of averaging of the signal during the measurement process which limits, but cannot remove, inherent fluctuation in the measurement. The fluctuation remaining limits the temperature resolution that can be achieved (Dicke 1946, Gabor 1950, Harvey 1963, Wait 1967). For all radiometer configurations this measurement noise is determined by three factors:

- (i) The system noise temperature, T_{sys} , the sum of the effective measured source noise temperature and the radiometer input noise temperature, including the effects of all signal circuit losses.
- (ii) The system noise power bandwidth, B_1 , before the high-frequency detector.
- (iii) The noise power bandwidth, B_2 , following high-frequency detection and including all signal processing before presentation of the measured temperature value.

This Gaussian noise component of the measured signal values takes the general form (Gabor 1950, Harvey 1963)

$$\Delta T_{\text{rms}} = CT_{\text{sys}} \sqrt{\frac{2B_2}{B_1}} \quad (1)$$

with C being a factor dependent on the radiometer configuration considered (see the appendix).

2.2. Loss dependence of signal equivalent temperature

In general, there must be some loss of signal power as a measured signal is transmitted through the microwave components of the input circuits of a radiometer and as reference temperature signals are transmitted through similar circuits. The loss in these circuits then contributes a proportional noise power dependent on the temperature of the loss region, introducing a systematic measurement error (Stelzried 1968, Wait and Nemoto 1968, Schwartz 1970). For the basic case of a uniform temperature circuit loss at T_α having an available power transmission ratio of α , the change in equivalent temperature caused by the circuit to a signal temperature T_S is $(1 - \alpha)(T_S - T_\alpha)$.

In a comparator radiometer temperature shifts of this nature will affect signals between both the source and the comparator switch and the reference noise source and the switch, and are the major cause of measurement drift. In the circuits following the comparator switch losses will have an indirect effect through their contribution to the overall radiometer noise temperature. The circuit temperatures can be measured by contact thermometry and with circuit loss calibrations can be used to apply corrections for the radiometric temperature changes occurring in the circuits (Stelzried 1968, Land 2001). The characteristic times associated with circuit temperature-dependent changes are

determined by the thermal capacities of the circuit elements or temperature sensors and the thermal impedances between them and local or ambient sources of varying temperature. Typical time constants are normally rather greater than the radiometer response time and are usually of the order of 10–1000 s.

2.3. Non-radiometric noise contributions

Microwave detection and post-detection amplifier noise can be included through the radiometer noise temperature (Lucas 1966, Land 1983), though it is usual to design the receiver system so that Gaussian and flicker noise contributions due to these sources are negligible. Further measurement noise or drift can, however, be introduced through any reference temperature term used to calculate the measured microwave temperature which is deduced from separate measurements using contact or other thermometry (figure A2 and (A.7)). Good design practice should, however, ensure that such noise contributions are small compared to the radiometric noise.

2.4. The Allan deviation for the analysis of microwave radiometer performance

The Allan variance is a two-sample variance formed by the average of the squared differences between successive values of a regularly measured quantity taken over sampling periods from the measuring interval up to half the maximum measurement time (Allan 1987, Levine 1999). In comparison with the commonly used standard variance, the Allan variance is based on measurement to measurement variation rather than on individual measurement to mean measurement variation. The Allan variance is defined so that it has the same value as the standard variance for measurement of Gaussian noise of uniform spectral power density. The Allan deviation is as for the standard deviation the square root of the variance, so that for N measurements of T_i and sampling period τ_0 (Barnes *et al* 1971, Allan 1987),

$$\sigma_y(\tau_0) = \sqrt{\frac{\sum_{i=1}^{N-1} (T_{i+1} - T_i)^2}{2(N-1)}}. \quad (2)$$

The sampling period is varied by averaging n adjacent values of T_i so that $\tau = n\tau_0$ and

$$\sigma_y(\tau) = \sqrt{\frac{\sum_{i=1}^{N-2n+1} (T_{i+2n} - 2T_{i+n} + T_i)^2}{2\tau^2(N-2n+1)}}. \quad (3)$$

The Allan deviation inherently provides a measure of the behaviour of the variability of a quantity as it is averaged over different measurement time periods, which allows it to directly quantify and to simply differentiate between different types of signal variation. The standard deviation does not provide such a direct way to distinguish types of noise or variation and thus to distinguish sources or causes of measurement variability (Allan 1987).

If different spectral noise components are assumed to be described by different spectral density power laws then examination of a log–log plot of Allan deviation versus sampling period allows different noise types to be distinguished by the slope of the plot in particular time regions and the magnitudes of these noise components to be determined (Lesage and Audoin 1973, Allan 1987, Levine 1999). The four types of signal variation of particular interest for microwave radiometry measurements are

- (i) The Gaussian noise inherent in the measured thermal signal, combined with the thermal noise generated in all of the component parts of the microwave amplification and detection circuits (Wells *et al* 1964, Land 1983). On the Allan deviation plot this noise type is associated with a region of slope -0.5 .
- (ii) Flicker noise generated in the active amplifying, detecting and temperature sensing components of the radiometer, usually having a noise corner where it merges with Gaussian noise in the 100 Hz–1 kHz region (Van der Ziel 1976, Cowley and Sorensen 1966). On the Allan deviation plot this form of noise contribution is shown as a region of slope 0.
- (iii) Random-walk noise, usually due to short-term changes in the temperature of microwave circuit losses and in amplifier gains that are not fully corrected for in the radiometer system. This type of variation is associated with an Allan deviation plot region of slope 0.5.
- (iv) Steady drift of the measurement values over times comparable to the data collection time, usually due to changes in the temperature of microwave circuit losses and in the determination of reference source temperatures. For linear drift the longer averaging time Allan deviation values, from (2), tend to $\frac{1}{\sqrt{2}}$ times the magnitude of the average gradient of the measurement data and the plot slope tends to 1.

The Allan deviation is defined so that for a white noise signal of uniform spectral power density extending well beyond the measurement sampling frequency it is equal to the standard deviation of that signal (Barnes *et al* 1971, Allan 1987). For many practical signal measurement systems and for microwave radiometry in particular the measured noise spectrum is low-pass limited and extends only to frequencies comparable to the signal sampling rate. It is common for the post-detection frequency response to provide a degree of pre-sampling anti-alias filtering which is followed by numerical filtering to obtain the wanted time response for the system, whilst providing sampling at a frequency close to the upper limit of the response (White 1989). This restricted spectrum then contains reduced signal components for the Allan deviation analysis at the shortest sampling times compared to an extended uniform spectrum and the Allan deviation estimates will be below the standard deviation value for the signal. If the power spectral density $S(f)$ of the noise signal is known, the correction to be applied to the Allan deviation to obtain the equivalent standard deviation can be found from the convolution of the noise spectrum with the effective Allan variance transfer function (Barnes *et al* 1971, Rutman 1974, Wiley 1977). For an averaging time τ the Allan variance for this restricted spectrum signal is

$$\sigma_y^2(\tau) = 2 \int_0^\infty S(f) \frac{\sin^2(\pi \tau f)}{(\pi \tau f)^2} \left\{ 1 - \frac{\sin^2(2\pi \tau f)}{4 \sin^2(\pi \tau f)} \right\} df. \quad (4)$$

For $S(f)$ determined by a gain-normalized measurement low-pass response $h(f)$, the Allan deviation (ADEV) to standard deviation (SDEV) ratio is then

$$\frac{\sigma_y(\tau)}{\sigma} = \sqrt{2 \int_0^\infty h^2(f) \frac{\sin^4(\pi \tau f)}{(\pi \tau f)^2} df \bigg/ \int_0^\infty h^2(f) df}. \quad (5)$$

The effect of the spectrum form is shown in figure 6 for a uniform noise spectrum cut off at unit frequency and for a Bessel fourth-order low-pass filtered spectrum of unit corner frequency.

For this work the microwave temperature and other data measured for the radiometers were analysed using the AlaVar 5 software package (Makdissi 2003) or a Matlab implementation. The AlaVar 5 software calculates the Allan deviation for doubling sampling periods across the measurement data set and also provides properly estimated upper and lower bounds for the ADEV values (Lesage and Audoin 1973, Makdissi 2003).

Figures 1–3 show examples of the types of data and Allan deviation plots used to determine the noise performance of the microwave radiometers. The measurement data are taken at 0.5 s intervals.

- (i) Figure 1 shows the output of a radiometer for low drift conditions. On the Allan deviation plot the slope of -0.5 marked identifies a region of predominantly Gaussian noise, with the maximum at 4 s averaging time of 51.6 ± 1.4 mK giving the ADEV value of this component. The roll-off of the ADEV value below 4 s is the convolution of the 3.3 s post-detection low-pass response (figure 5) with the effective filter response of the analysis, giving an ADEV value of 0.74 of the SDEV value (figure 6).
- (ii) Figure 2 shows measurement data from the same radiometer in the presence of induced near-linear drift. The Allan deviation plot changes from the Gaussian region slope -0.5 to the drift region slope of 1 above about 100 s. With (iv) above, the mean Allan deviation derived drift above 200 s averaging time of $106 \mu\text{K s}^{-1}$ is equal to the slope of the linear fit to the data of $106.5 \mu\text{K s}^{-1}$.
- (iii) Figure 3 shows an example of radiometer measurement data and the corresponding Allan deviation plot when there is significant quasi-random-walk variation present. Here this produces an ADEV plot region of slope 0.5 above about 200 s averaging time.

3. Application of Allan deviation analysis to microwave radiometer measurements

3.1. Measurement of radiometer noise performance

The noise performances of two microwave radiometers have been measured and compared with noise models. One radiometer is of the two-reference configuration and the other is of the input balancing or Dicke type (appendix). The noise performances were measured with an attenuated noise diode source (figure 4) or a water-bath immersed antenna providing the variable temperature signals. The noise diode source has been found to be very stable in use (Randa 2001) and by providing switched and variable equivalent temperature signals to be particularly convenient for this type of investigation. The adjustable attenuation of the noise diode source allowed the measured temperature to be set to provide two-reference radiometer signal component ratios over a range $\geq \pm 2$ (A.7, A.8), which for the reference temperatures used for this radiometer ($T_{r1} \approx 26^\circ\text{C}$ and $T_{r2} \approx 78^\circ\text{C}$) corresponded to a source equivalent temperature range of approximately 1°C – 104°C .

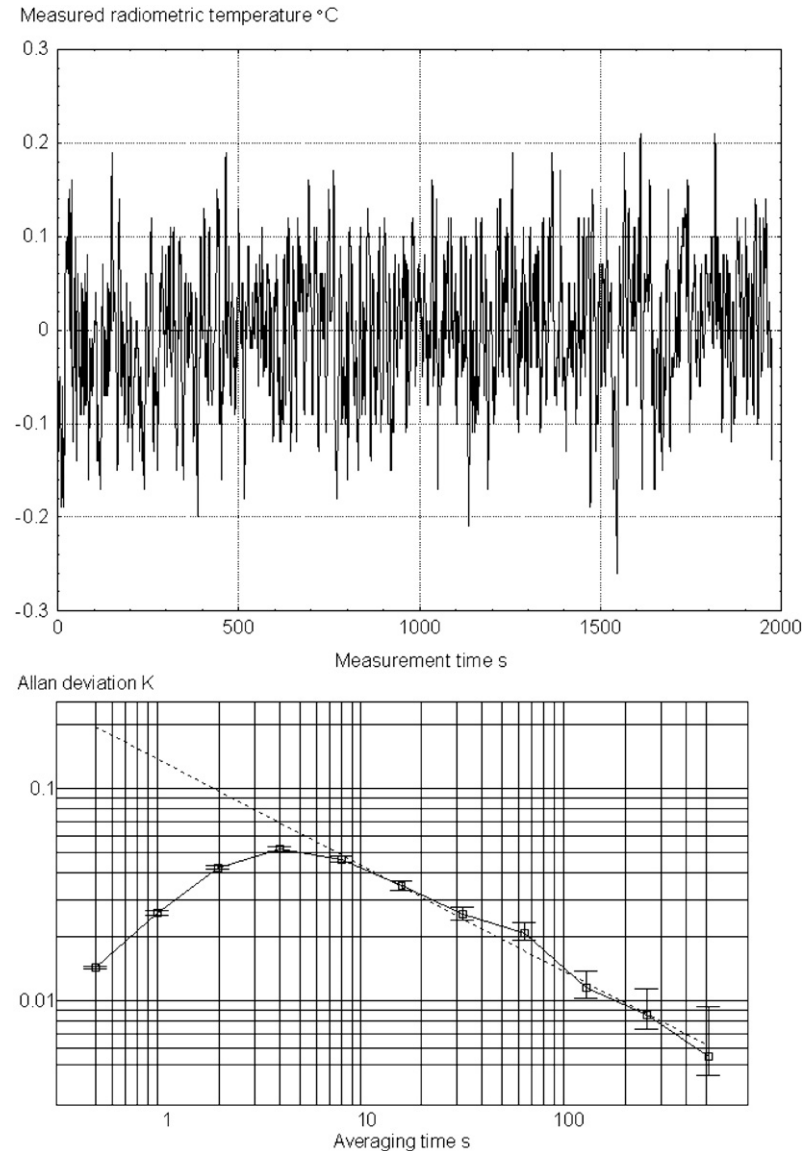


Figure 1. Low drift microwave temperature data sampled at 0.5 s intervals from a 3.0–3.5 GHz two-reference type radiometer with a measurement response time of 3.3 s and the Allan deviation (ADEV) plot derived from the data. The line of slope -0.5 (marked) indicates the region of uniform spectral density Gaussian noise. The roll-off of the ADEV value below 4 s is the convolution of the 3.3 s post-detection low-pass response with the effective filter response of the 0.5 s sampling analysis, giving a maximum ADEV value of 0.74 of the SDEV value from figure 6.

3.2. Two-reference radiometer performance

The radiometer used for this section of the investigation is a 3.0–3.5 GHz single channel two-reference radiometer for industrial and medical use over the range -20°C – 120°C (Land 2001). A dual PIN-diode, dual circulator input circuit switches the source and two reference loads at 1 kHz. The detected microwave signal is synchronously demodulated to obtain the in-phase and quadrature components of the switched signal which are then numerically processed to give the source microwave temperature (appendix A.4). The post-detection response of this radiometer is determined by a second-order near critically damped Sallen-Key low-pass filter of 0.154 Hz corner frequency followed by Hamming window numerical filtering to give the overall transient and transformed frequency

responses of figure 5. The overall post-detection noise power bandwidth is 0.11 Hz.

Figure 6 shows the effective response for Allan deviation analysis of the radiometer low-pass filtering with the numerical sampling frequency of 2 Hz. The ADEV/SDEV ratios for the Gaussian noise ADEV region were obtained from low drift radiometer measurement data (figure 1), taking SDEV values from the residual values left after stripping off cubic drift fitting. Investigation of higher order, cyclical and heavily smoothed function stripping showed that *ad hoc* cubic function stripping gave similarly minimal SDEV values for the data used. At the ADC sampling frequency of 2 Hz, however, there is still significant gain through the preceding second-order filter which allows generation of ADC alias products to give an effective enhancement to the high-frequency end of the noise

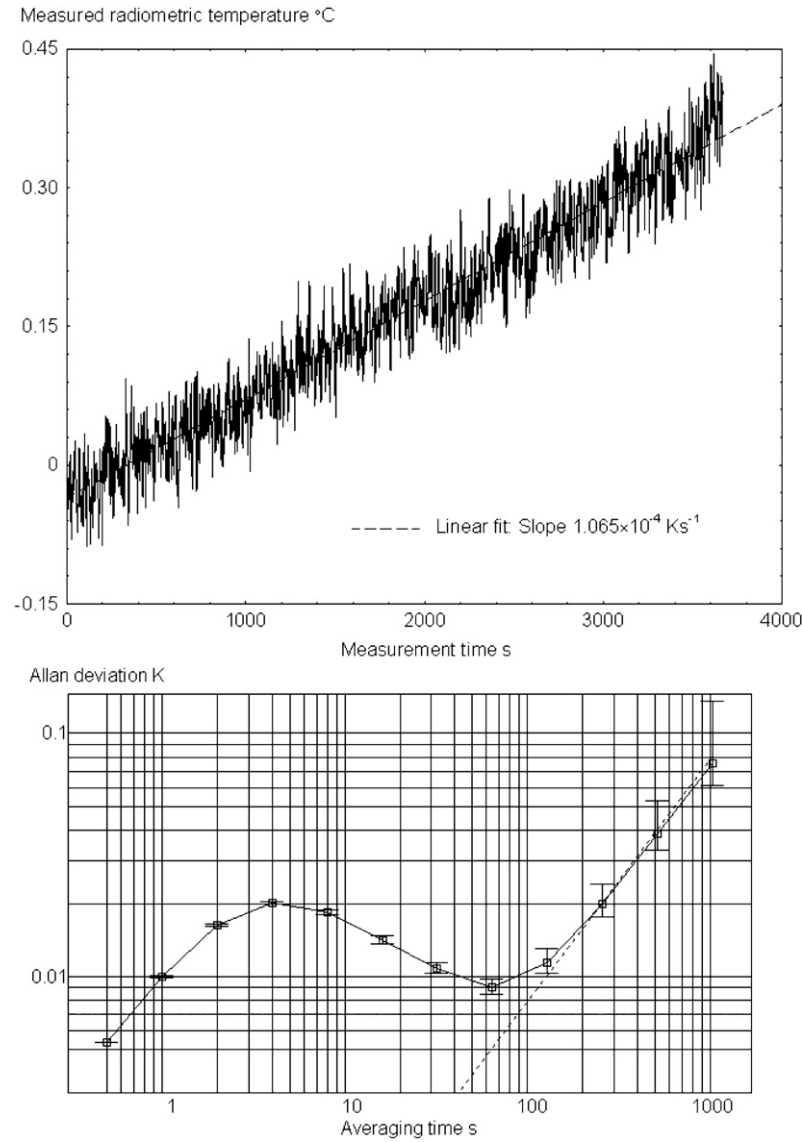


Figure 2. Microwave radiometric temperature data as for figure 1, but with the source allowed to drift, and the corresponding Allan deviation plot. A linear fit to the data gives a mean rate of drift over the measurement time of $106.5 \mu\text{K s}^{-1}$. On the Allan deviation plot the slope 1 region (marked) shows a drift component of $106 \mu\text{K s}^{-1}$, merging with Gaussian noise which predominates below 60 s averaging time.

spectrum (White 1989) and which raises the ADEV/SDEV ratio values for the shortest averaging times.

The overall Gaussian noise behaviour for the two-reference radiometer was obtained from Allan deviation data, corrected as above, for temperatures corresponding to signal component ratios over the range ± 2 (A.7). Using hot- and cold-load measurements to obtain the radiometer noise temperature ($410 \pm 20 \text{ K}$) (Engen 1973), conventional swept-frequency response measurement with (A.1) to obtain the pre-detection noise power bandwidth ($420 \pm 10 \text{ MHz}$), and the post-detection noise power bandwidth as above, the noise behaviour was modelled as (A.9). The comparison of measured and modelled noise behaviour in figure 7 shows good agreement of both the form of the variation and of the absolute values.

Allan deviation plots were also made of the directly measured reference load temperatures that provide the terms $\frac{T_{r1}-T_{r2}}{2}$ and $\frac{T_{r1}+T_{r2}}{2}$ used with the signal component ratio to

calculate the measured source temperature (A.7). These showed response corrected quasi-Gaussian noise levels of about 3 mK (figure 8), which is of the order of the ADC quantization noise expected for this temperature sensing. This noise is uncorrelated with the microwave radiometric noise and so noise from these terms will contribute 0.1 mK or less to the Dicke level measurement noise of 33 mK.

3.3. Variable reference temperature input-balancing radiometer

The radiometer used for this part of the investigation is one 3.4–3.8 GHz channel of a five-band Dicke configuration system developed for a specialized medical application (Hand *et al* 2001). A PIN-diode and circulator input circuit switches between the source and reference load at 1 kHz. The detected microwave signal is synchronously demodulated to obtain the switched signal component which is then numerically

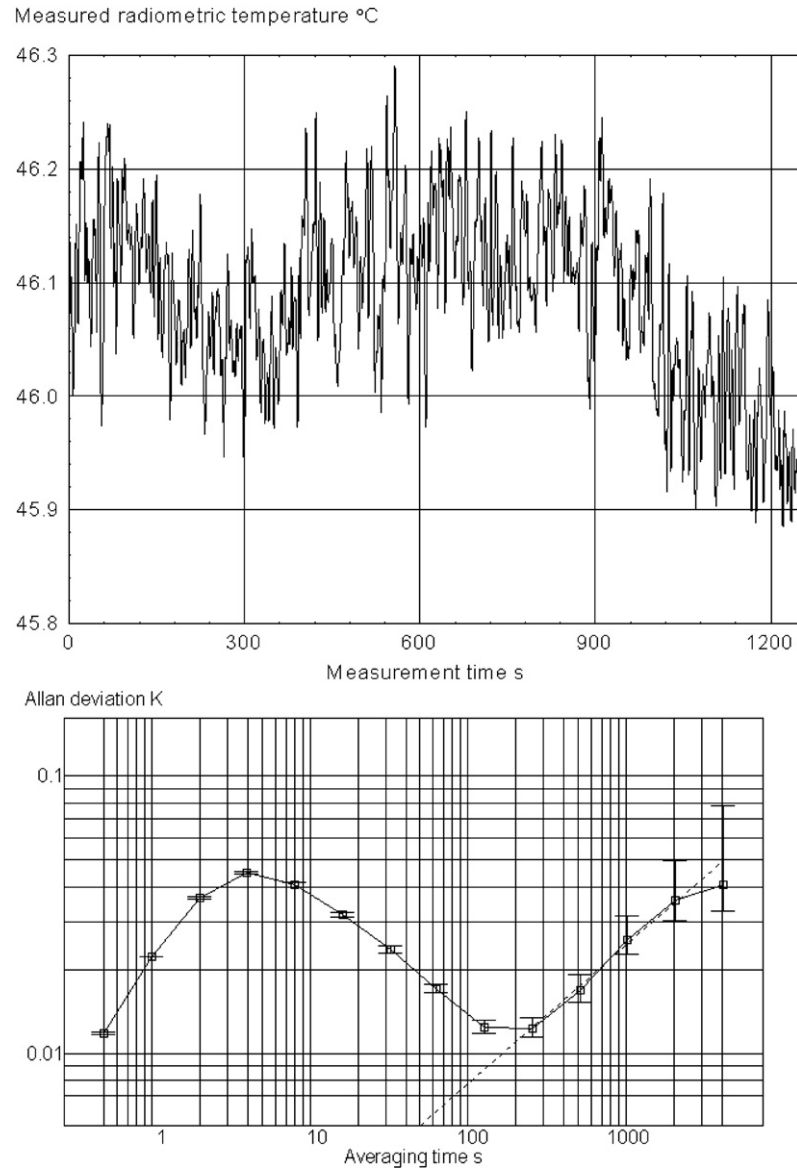


Figure 3. Microwave radiometric temperature data measured as for figure 1, but showing a significant quasi-random-walk component, and the corresponding Allan deviation plot. The Allan deviation plot shows a region of slope 0.5 (marked) above about 200 s averaging time produced by the quasi-random-walk components in the data, and the predominance of Gaussian noise below 100 s averaging time.

processed to control the reference temperature to null this source-reference difference signal (appendix A.3). The reference load source is a coaxial $50\ \Omega$ termination which has its temperature PID controlled by a Peltier device and measured with a thermistor. When the null condition at the switch is met, the temperature of the source being measured is equal to that of the reference noise source with appropriate calibration to allow for the losses of the input and reference load signal paths.

Figure 9 shows the time response of the radiometer T_{ref} value to a near step change in the source temperature between water baths at $30.3\ ^\circ\text{C}$ and $40.8\ ^\circ\text{C}$, and the modelled behaviour as (A.6). For $\tau = 60\ \text{s}$ and $g = 0.27$, the effective time constant is 191 s. The positive transients on the measured values of T_{ref} are due to the Peltier device PID control loop overshooting for short timescales and this is neglected in the analysis.

Sets of microwave temperatures sampled at 1.2 s intervals were recorded over approximately 1 h periods and Allan deviation plots generated from the data. Figure 10 shows a typical plot for a water-bath temperature of $40\ ^\circ\text{C}$. For this particular radiometer configuration the post-detection noise bandwidth is very low, $\sim 1.3 \times 10^{-3}\ \text{Hz}$ (from figure 8), giving an expected Gaussian noise contribution from the system noise temperature of less than 3 mK (A.4).

A simple model has been developed to simulate the Allan deviation plot for this radiometer system. The radiometer is considered functionally equivalent to a linear measurement system in which an input signal is transformed into an output signal by applying a filter function. Two sources of noise are added into the measurement system: pre-filter noise inherent in the input signal and reference sensor noise added at the post-filter stage (figure A2(b)). The procedure to simulate the Allan deviation plot is to calculate the power spectral density

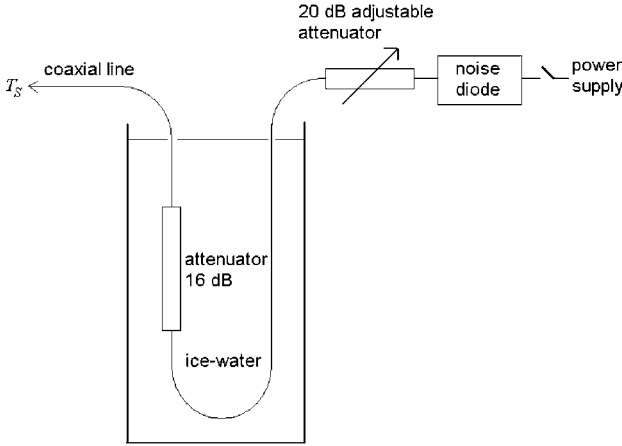


Figure 4. Continuously and rapidly adjustable noise source for radiometer noise performance measurement providing equivalent noise temperatures of 0.7 °C–120 °C.

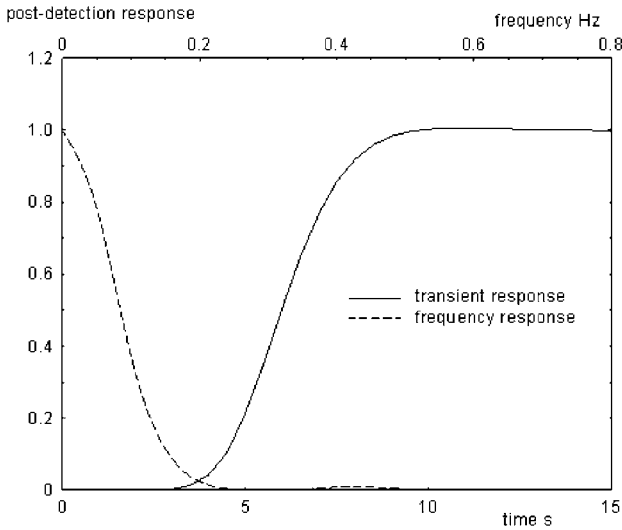


Figure 5. Transient response and transformed frequency response of the two-reference radiometer system determining the post-detection noise power bandwidth.

function for the equivalent measurement system and then compute the Allan variance by applying the transform given by equation (4). The parameters defining the pre- and post-filter noise levels are manually adjusted until the simulated and experimental Allan deviation plots agree with one another to within the experimental uncertainties. The filter function used in the simulation is the iterative function (A.5) transformed into the frequency domain. The pre-filter and post-filter noise sources are found to be dominated by flicker ($1/f$) noise in the water-bath source temperature and Gaussian noise in the reference sensing thermistor, respectively.

Figure 10 shows measured and simulated Allan deviation plots, for which the simulation noise parameters have been adjusted until they agree within experimental uncertainties. The pre-filter flicker ($1/f$) noise parameter (h_{-1}) is 0.0005 K² and the post-filter Gaussian noise parameter (h_0) is 0.00016 K² Hz⁻¹. The noise parameters are defined by $S(f) = h_0$ and $S(f) = \frac{h_{-1}}{f}$ for Gaussian and flicker noise respectively, where $S(f)$ is the power spectral density

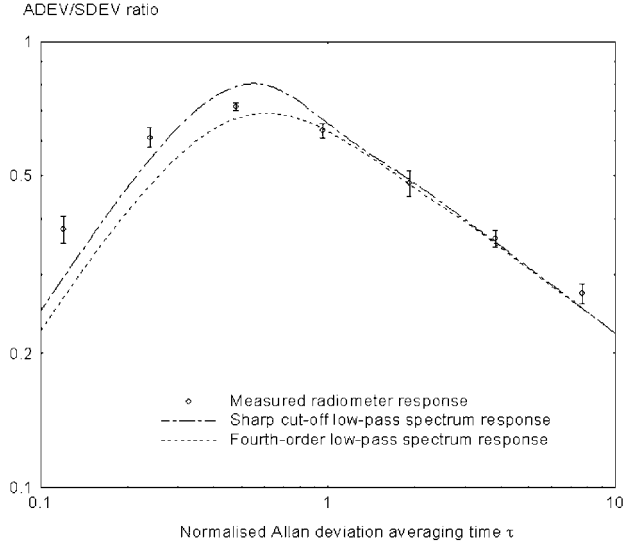


Figure 6. Two-reference radiometer measured Allan deviation to standard deviation ratio (ADEV/SDEV) compared with behaviour expected for unit-frequency sharp cut-off response and a fourth-order response close to the overall radiometer response.

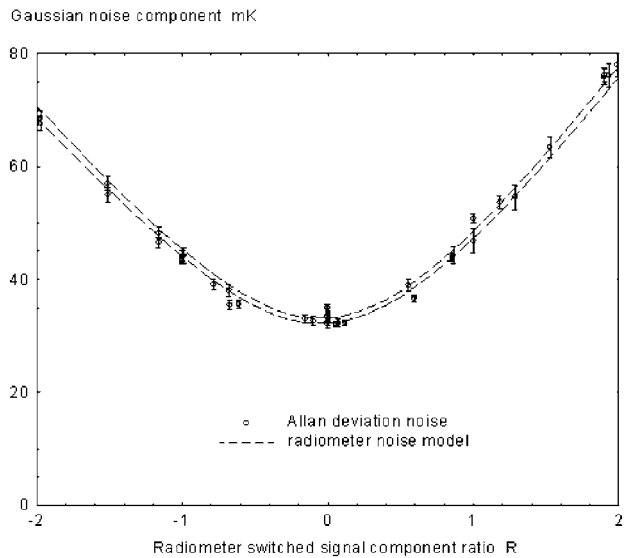


Figure 7. Comparison of measured and modelled radiometric measurement noise for a 3.0–3.5 GHz two-reference radiometer using reference temperatures of 26 °C and 78 °C and measuring source temperatures from 0 °C to 104 °C (A.7). The modelling is as (A.9) with a pre-detection bandwidth of 420 ± 10 MHz, post-detection response of figure 5, and a radiometer input noise temperature of 410 ± 20 K. The modelled equivalent Dicke configuration measurement noise is 33 ± 1.5 mK ($R = 0$ condition).

function. The standard deviation (σ) of the Gaussian noise is $\sqrt{h_0} = 0.0126$ K for 1 s averaging time.

The Allan deviation plots can thus be interpreted as showing three measurement variation regions:

- Between 1 s and approximately 5 s where the Gaussian noise from the reference load sensor after the post-detection filtering is dominant.
- Between about 5 s and 500 s where the source related noise seen through the roll-off of the post-detection filtering of the iterative control process is dominant.

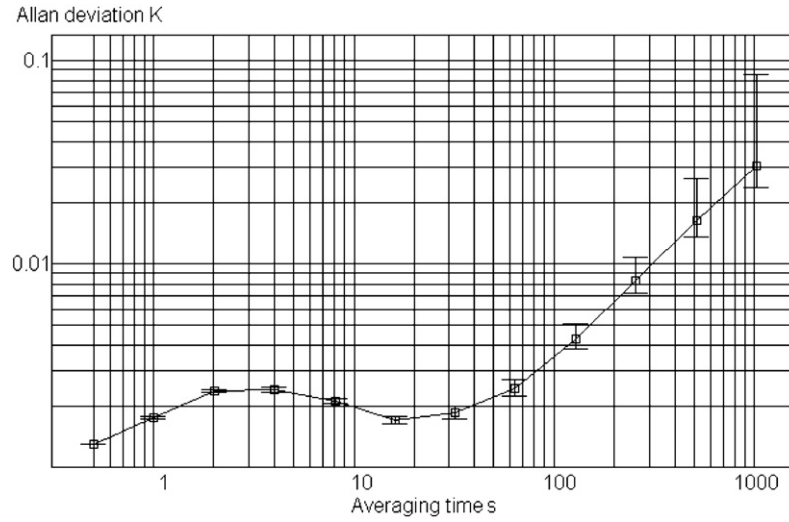


Figure 8. Allan deviation plot of thermistor derived reference temperature differences $\frac{T_{r1}-T_{r2}}{2}$ for the two-reference radiometer. Below about 20 s averaging time the main noise component is quasi-Gaussian noise equivalent to 3 mK SDEV. Above about 50 s averaging time the drift in the reference load temperature becomes dominant.

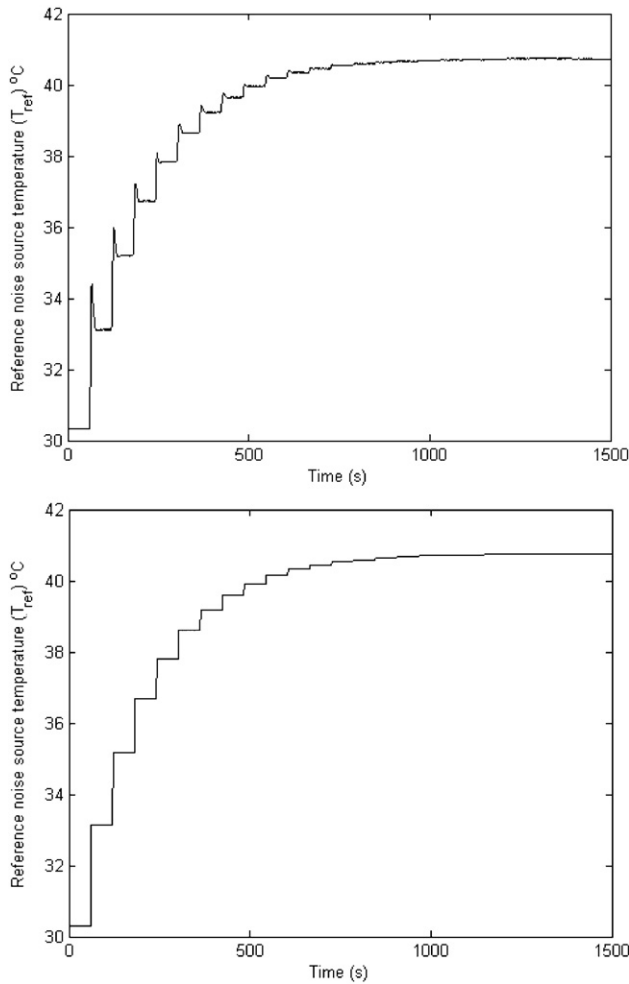


Figure 9. Measured (upper) and simulated (lower) responses of the input balancing radiometer to a step change in source temperature from 30.3 °C to 40.8 °C. The radiometer behaves as the modelled system of figure A4 with the response of (A.6) for delay $\tau = 60$ s and gain $g = 0.27$ giving an effective time constant of 191 s.

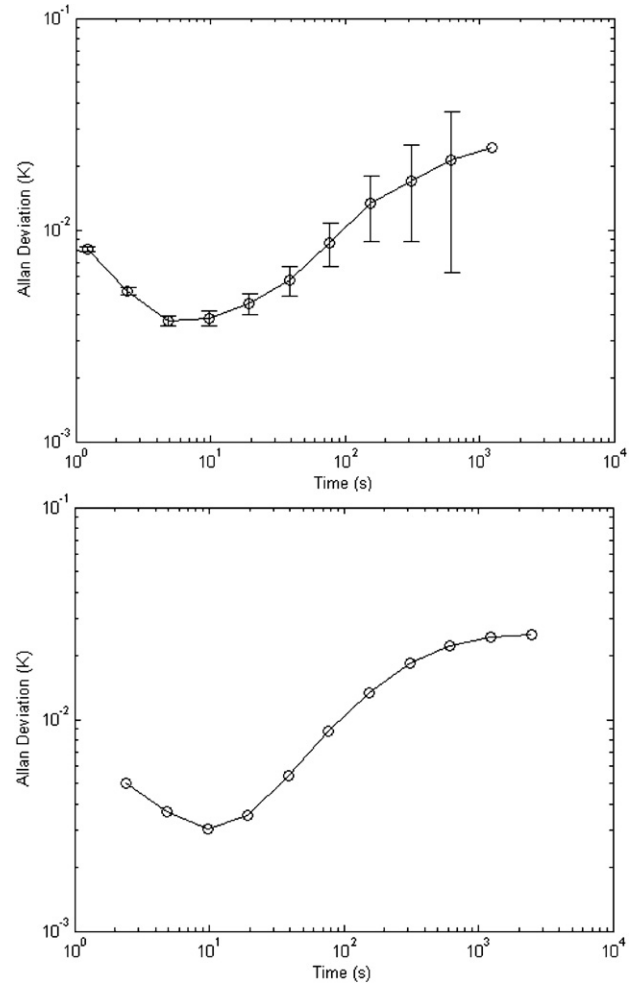


Figure 10. Comparison of measured (upper) and simulated as figure A4 Allan deviation behaviour for the input nulling radiometer. For these measurement conditions, the pre-filter flicker ($1/f$) noise parameter (h_{-1}) is 0.0005 K² and the post-filter Gaussian noise parameter (h_0) is 0.000 16 K² Hz⁻¹.

- (iii) Above about 500 s where the random fluctuations in the water-bath temperature or temperature variations in the antenna cable losses dominate.

4. Conclusions

Allan deviation analysis can identify and provide excellent differentiation between regions of Gaussian noise, flicker noise and drift in microwave radiometric temperature and similar measurements. For comparisons with Gaussian noise values expressed in terms of the standard deviation, or for comparisons between instruments, the spectrum of the analysed noise signal must be known and the Allan deviation corrected for the convolution of the spectrum with the response of the analysis sampling. With this factor the Allan deviation provides an easily obtained and universally applicable measure of the Gaussian noise component of a signal which avoids the difficulty of determining the proper standard deviation measure in the presence of drift. The Allan deviation plot shows clearly the relative importance of noise and drift in measurement data and the time regions over which these variations are dominant. The Allan deviation analysis has been applied to investigate the noise performance of two very different microwave radiometry instruments and has both guided and accurately confirmed the noise modelling developed for these systems.

Acknowledgments

The investigation of this application of the Allan deviation analysis technique has been supported by the University of Glasgow, the National Physical Laboratory, and Hammersmith Hospitals NHS Trust. The development of the microwave radiometers used for the investigation was supported by the UK Engineering and Physical Sciences Research Council, the Garfield Weston Foundation, Imperial College London, Northern Foods plc, Loma Engineering Ltd and the University of Glasgow

Appendix: Measurement noise in microwave radiometers

A.1. Total power radiometer

In a total power radiometer (figure A1(a)) the microwave signal input is continuously connected to the thermal noise source at equivalent temperature T_S which is to be measured. In general there will be some impedance mismatch between the source and radiometer that can be represented by a power reflection coefficient ρ (Wait and Nemoto 1968). The pre-detection microwave noise power bandwidth is B_1 and the average power gain is G , and a post-detection noise power bandwidth is B_2 .

The detected noise signal, referenced to the radiometer input, is the system noise temperature T_{sys} due to the source T_S plus the radiometer noise equivalent temperature T_{rad} . (Adler *et al* 1963). The noise power into the detector is $Gk(T_S - \rho(T_S - T_{\text{rad}}))B_1$, with k being Boltzmann's constant, which for a square-law microwave detector having an output voltage proportional to input microwave power P_{in}

of $V = KP_{\text{in}}$, will give an average detector output $\bar{V} = KGk(T_S - \rho(T_S - T_{\text{rad}}))B_1$. This measure of the source temperature is dependent on the uncontrolled quantities of gain G , amplifier noise temperature T_{rad} and source reflection coefficient ρ .

For a matched source and a uniform pre-detection noise spectral power density $w_0 = GkT_{\text{sys}}$, $\bar{V} = KGkT_{\text{sys}}B_1$ and $\bar{V} = Kw_0B_1$. The multiplicative action of the square-law detector transforms this spectral density to the band-limited triangular output spectral density distribution $w_2(f) = 2w_0^2(B_1 - f)$ (Van der Ziel 1955, Meredith *et al* 1964, Lucas 1966), which for low post-detection frequencies has spectral power density $w_2(0) = 2w_0^2B_1$. The noise power bandwidth can be defined to be both consistent with this relationship and independent of the form of the gain-frequency response $G(f)$ by (Kittel 1977, Roberts and Blalock 1985)

$$B_1 = \left[\int_{-\infty}^{\infty} |G(f)|^2 df \right]^2 / \int_{-\infty}^{\infty} |G(f)|^4 df. \quad (\text{A.1})$$

The detector output signal is measured through post-detection circuits having a noise power bandwidth B_2 defined for the complete post-detection system frequency response $(h(f))$. The mean-square deviation of V from its mean value $\bar{V}$ for $f \ll B_1$ is then $\overline{(V - \bar{V})^2} = K^2 w_2(0) B_2 = K^2 2w_0^2 B_1 B_2$.

Taking the system to be calibrated so that $\bar{V} = cT_{\text{sys}} = c(T_S + T_{\text{rad}})$ and $\frac{\partial V}{\partial T_S} = c$, the root-mean-square equivalent temperature fluctuation of this signal is

$$\Delta T_{\text{rms}} = \sqrt{\frac{\overline{(V - \bar{V})^2}}{c^2}} = \sqrt{\frac{T_{\text{sys}}^2 K^2 2w_0^2 B_1 B_2}{(Kw_0 B_1)^2}} = T_{\text{sys}} \sqrt{\frac{2B_2}{B_1}}. \quad (\text{A.2})$$

This is the minimum possible measurement equivalent temperature fluctuation, the 'Gabor limit', applying to all microwave radiometers (Gabor 1950, Harvey 1963). The measurement response-time is determined by the form of the overall system post-detection frequency response that defines the noise power bandwidth B_2 , this response including the effects of any computational post-detection signal processing. With an optimized transient response the measurement time $\tau_{\text{opt}} \approx 0.35/B_2$ (Terman 1955, Land 1983) giving a temperature resolution to response-time relationship

$$\Delta T_{\text{rms}} \sqrt{\tau_{\text{opt}}} \approx 0.84 \frac{T_{\text{sys}}}{\sqrt{B_1}}. \quad (\text{A.3})$$

A.2. Single reference comparator radiometer

The dependence of total power radiometer measurements on poorly controlled system properties is removed or reduced in the single-reference comparator or Dicke radiometer configuration (Dicke 1946, Harvey 1963, Wait 1967). Here the input to the radiometer microwave amplifier is continuously switched between the measured source and a known temperature reference source T_{ref} . The output from the detector, containing the switched microwave signal component, is then synchronously demodulated at the switching frequency (figure A1(b)).

The demodulated difference signal is $\Delta \bar{V}_s = KG(T_S - \rho(T_S - T_{\text{rad}}) - T_{\text{ref}})B_1$. If the equivalent temperature of the

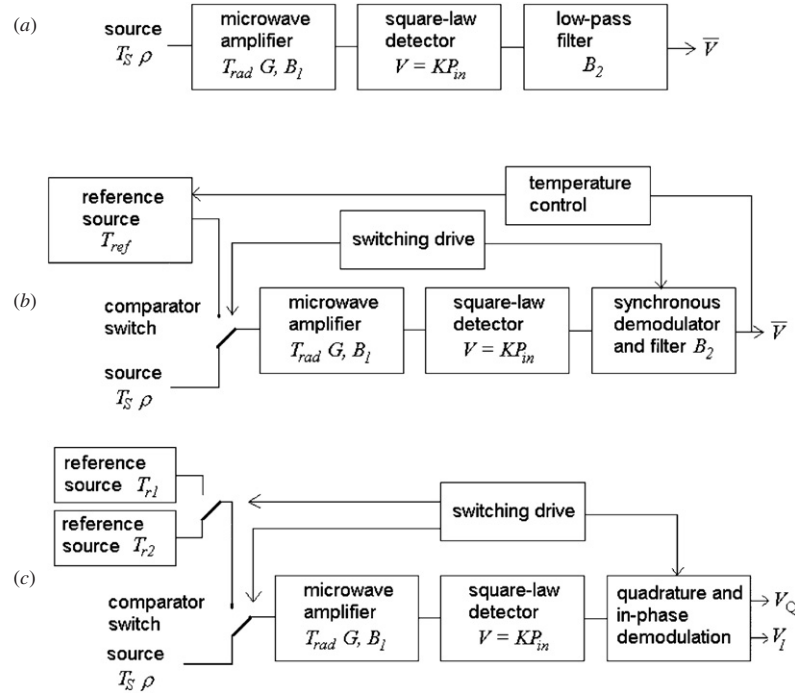


Figure A1. (a) Total power radiometer configuration. (b) Comparator or Dicke configuration radiometer applicable to the input nulling radiometer. (c) Two-reference radiometer configuration.

reference source is adjusted so that $\Delta \bar{V}_s = 0$, then $T_{\text{ref}} = T_S - \rho(T_S - T_{\text{rad}})$, the dependence on gain is removed, and for a matched source the source and reference equivalent temperatures are equal (Ludeke *et al* 1978).

Compared with the total power radiometer configuration the calibration of the system is now $\frac{\partial \bar{V}}{\partial T_S} = \frac{\epsilon}{2}$ since the average source signal power is half that for the total power radiometer, and for the usual operating condition of equal source and reference switched times the average system temperature over a switching cycle is $T_{\text{sys}} = \frac{T_S + T_{\text{ref}}}{2} + T_{\text{rad}}$. This gives the equivalent root-mean-square temperature fluctuation for the Dicke radiometer configuration

$$\Delta T_D = 2 \left(\frac{T_S + T_{\text{ref}}}{2} + T_{\text{rad}} \right) \sqrt{\frac{2B_2}{B_1}} = 2T_{\text{sys}} \sqrt{\frac{2B_2}{B_1}}. \quad (\text{A.4})$$

A.3. Input nulling radiometer

In the reference balancing radiometer considered here the physical temperature of the reference load is controlled to achieve input null balance according to the flow chart of figure A2(a) (Hand *et al* 2001). The control computer periodically reads the reference temperature sensor and the synchronous demodulator output and applies a new temperature set point to the reference source according to the relationship

$$T_{\text{ref}}(t + \tau) = T_{\text{ref}}(t) - g \langle T_{\text{ref}}(t) - T_S(t) \rangle, \quad (\text{A.5})$$

where t is the time interval between adjustments, T_S is the source temperature, $T_{\text{ref}}(t)$ is the set point of the reference noise temperature at time t , and g is the system gain. Provided $0 < g < 1$, the iterative process will converge to $T_{\text{ref}} = T_S$ when the source temperature is taken as equalling the reference sensor value.

For the first time interval τ , $\Delta T_1 = T_{\text{ref}}(\tau) - T_S = (T_{\text{ref}}(0) - T_S)(1 - g) = \Delta T_0(1 - g)$, and for the n th time interval $\Delta T_n = \Delta T_0(1 - g)^n$ with $0 < g < 1$. Setting $\Delta T_n = \frac{\Delta T_0}{e}$ then

$$n = \frac{-1}{\ln(1 - g)} \quad \text{and} \quad \tau_R = n\tau = \frac{-\tau}{\ln(1 - g)}. \quad (\text{A.6})$$

This iterative process can thus be considered equivalent to a post detection low-pass filter with an effective time constant of τ_R .

Using this relationship Allan deviation noise plots were generated by the simulation procedure of figure A2(b) where noise is added to the system at two points as

- (i) pre-iteration filter noise h_1 comprising essentially flicker ($1/f$) or random drift variations due to temperature changes of source elements of the system, and
- (ii) post-filter noise h_0 comprising Gaussian (white) noise on the reference sensor value T_{ref} .

The pre-iteration noise is low-pass filtered by the system nulling response (figure 9 and (A.6)) of approximately -6 dB per octave to produce the dominant system noise behaviour of figure 10.

A.4. Two-reference radiometer

The two-reference radiometer is related to the single-reference radiometer in configuration but takes the developed form of figure A1(c) and uses fixed temperature references (Land 2001). It is operated so that the reference temperature is switched between two values T_{r1} and T_{r2} for equal times within each half of the input switching cycle. Two post-detection

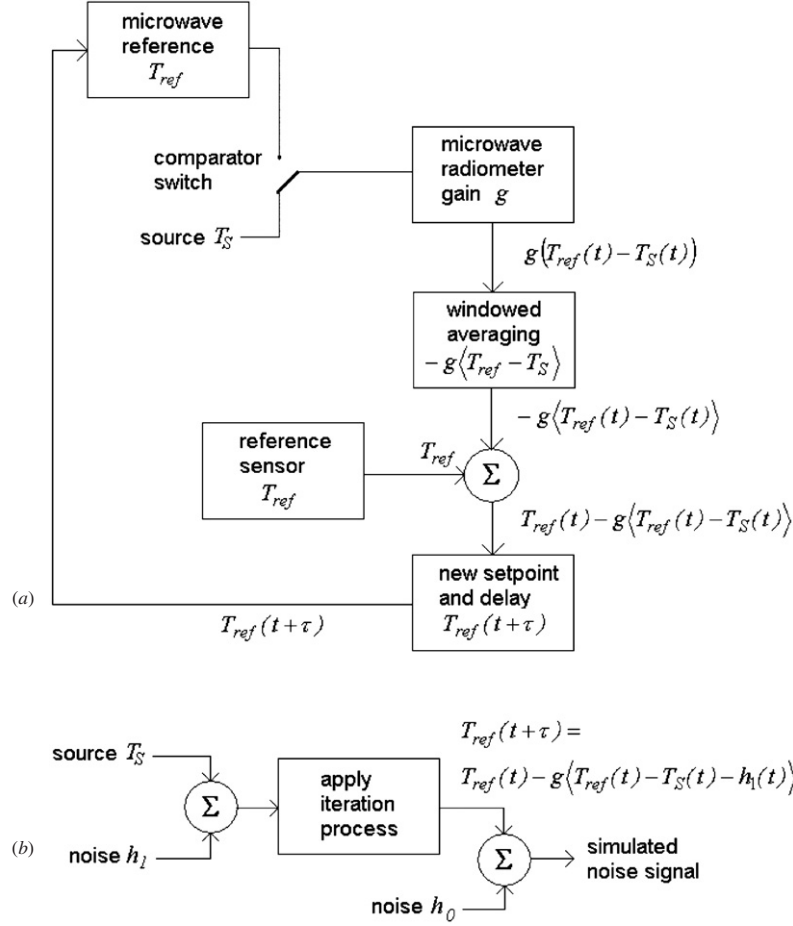


Figure A2. (a) Flow diagram for the complete measurement path of the input-balancing radiometer, showing the iterative process used to achieve null balance of $T_{ref} = T_S$. (b) Simulation of noise behaviour in the input-balancing radiometer for the generation of Allan deviation plots with the flicker noise spectrum h_1 added to the source signal and the Gaussian noise spectrum h_0 added to the reference sensor signal.

synchronous detectors extract the in-phase and quadrature-phase components of the detected microwave signal, V_I and V_Q .

Through the action of the two switches the four phases of the switching cycle connect the amplifier input to provide four noise temperature levels of

$$\begin{aligned} T_1 &= T_{r1} + T_{rad} \\ T_2 &= T_{r2} + T_{rad} \\ T_3 &= \rho T_{r1} + (1 - \rho) T_S + T_{rad} \\ T_4 &= \rho T_{r2} + (1 - \rho) T_S + T_{rad}. \end{aligned}$$

After detection the signal generated by the switching cycle is synchronously demodulated to provide in-phase and quadrature-phase components

$$\begin{aligned} V_I &= (T_2 + T_3) - (T_1 + T_4) \\ V_Q &= (T_1 + T_2) - (T_3 + T_4). \end{aligned}$$

With R being the ratio of these components, the source equivalent temperature is given by

$$T_S = R \left(\frac{T_{r1} - T_{r2}}{2} \right) + \left(\frac{T_{r1} + T_{r2}}{2} \right). \quad (\text{A.7})$$

The radiometer is calibrated to provide reference temperature terms T_{r1} and T_{r2} derived from contact thermometry, so that these terms do not contribute to the microwave signal-dependent noise of the system.

For microwave noise uncorrelated over the phases of the switching cycle, the mean-square variation of the ratio R is

$$\begin{aligned} \delta R^2 &= \frac{2B_2}{B_1(T_{r1} - T_{r2})^2} \\ &\times [(R^2 + 1)((T_{r1} + T_{rad})^2 + (T_{r2} + T_{rad})^2 + 2(T_S + T_{rad})^2) \\ &+ 2R(T_{r1} - T_{r2})(T_{r1} + T_{r2} + 2T_{rad})]. \end{aligned} \quad (\text{A.8})$$

With $\delta T_S = \frac{T_{r1} - T_{r2}}{2} \delta R$, the equivalent temperature fluctuation is

$$\begin{aligned} \Delta T_{2rms} &= \sqrt{\frac{2B_2}{B_1} [(R^2 + 1)((T_{r1} + T_{rad})^2 + (T_{r2} + T_{rad})^2 \\ &+ 2(T_S + T_{rad})^2 + 2R(T_{r1} - T_{r2})(T_{r1} + T_{r2} + 2T_{rad})]^{1/2}}. \end{aligned} \quad (\text{A.9})$$

If the individual temperatures of the switching phases are close to the mean system noise temperature this can be simplified to

$$\begin{aligned} \Delta T_{2rms} &\approx 2 \sqrt{\frac{2B_2}{B_1} \left(\left(\frac{T_{r1} - T_{r2}}{4} \right) R + \frac{T_{r1} + T_{r2}}{2} + T_{rad} \right) \sqrt{R^2 + 1}}. \end{aligned} \quad (\text{A.10})$$

This shows that in comparison with the single-reference radiometer the measurement noise here has a direct dependence on a signal component ratio term of $\sqrt{R^2 + 1}$.

References

- Adler R, Engelbrecht R S, Harrison S W, Haus H A, Lebenbaum M T and Munford W W 1963 Description of the noise performance of amplifiers and receiving systems *Proc. IEEE* **51** 436–42
- Allan D W 1966 Statistics of atomic frequency standards *Proc. IEEE* **54** 221–30
- Allan D W 1987 Should classical variance be used as a basic measure in standards metrology? *IEEE Trans. Instrum. Meas.* **36** 646–54
- Barnes J A *et al* 1971 Characterization of frequency stability *IEEE Trans. Instrum. Meas.* **20** 105–20
- Bardati F, Marrocco G and Tognolatti P 2004 Time-dependent microwave radiometry for the measurement of temperatures in medical applications *IEEE Trans. Microw. Theor. Tech.* **52** 1917–24
- Carr K L, El-Mahdi A M and Shaeffer J 1981 Dual-mode microwave system to enhance early detection of cancer *IEEE Trans. Microw. Theor. Tech.* **29** 256–60
- Chive M, Plancot M, Giaux G and Prevost B 1984 Microwave hyperthermia controlled by microwave radiometry: technical aspects and first clinical results *J. Microw. Power* **19** 233–41
- Cowley A M and Sorensen H O 1966 Quantitative comparison of solid-state microwave detectors *IEEE Trans. Microw. Theor. Tech.* **14** 588–602
- Dicke R H 1946 The measurement of thermal radiation at microwave frequencies *Rev. Sci. Instrum.* **17** 268–75
- Engen G F 1973 Mismatch considerations in evaluating amplifier noise performance *IEEE Trans. Instrum. Meas.* **22** 274–9
- Foster K R and Cheever E A 1992 Microwave radiometry in biomedicine: a reappraisal *Bioelectromagnetics* **13** 567–79
- Gabor D 1950 Communication theory and physics *Phil. Mag.* **41** 1161–87
- Goodberlet M A and Mead J B 2006 Two-load radiometer precision and accuracy *IEEE Trans. Geosci. Remote Sensing* **44** 58–66
- Hand J W, Van Leeuwen G M J, Mizushima S, Van de Kamer J B, Maruyama K, Sugiura T, Azzopardi D V and Edwards A D 2001 Monitoring of deep brain temperature in infants using multi-frequency microwave radiometry and thermal modelling *Phys. Med. Biol.* **46** 1885–904
- Harvey A F 1963 *Microwave Engineering* (London: Academic) pp 773–82
- Huntley L 1988 An investigation into the role of the Allan Variance in establishing a reference standard of voltage *Precision Electromagnetic Measurements CPEM 88 Digest* pp 15–6
- Kittel P 1977 Comment on the equivalent noise bandwidth approximation *Rev. Sci. Instrum.* **48** 1214–5
- Land D V 1983 Radiometer receivers for microwave thermography *Microw. J.* **26** 196–201
- Land D V 1987 A clinical microwave thermography system *Proc. IEE* **134A** 193–200
- Land D V 2001 An efficient, accurate and robust radiometer configuration for microwave temperature measurement for industrial and medical applications *J. Microw. Power Electromag. Energy* **36** 139–54
- Leroy Y, Mamouni A, Van De Velde J C, Bocquet B and Dujardin B 1987 Microwave radiometry for non-invasive thermometry *Automedica* **8** 181–202
- Leroy Y, Bocquet B and Mamouni A 1998 Non-invasive microwave radiometry thermometry *Physiol. Meas.* **19** 127–48
- Lesage P and Audoin C 1973 Characterization of frequency stability: uncertainty due to finite number of measurements *IEEE Trans. Instrum. Meas.* **22** 157–61
- Levine J 1999 Introduction to time and frequency metrology *Rev. Sci. Instrum.* **70** 2567–95
- Lucas W J 1966 Tangential sensitivity of a detector-video system with r.f. preamplification *Proc. IEE* **113** 1321–30
- Ludeke K M, Schiek B and Koehler J 1978 Radiation balance microwave thermograph for industrial and medical applications *Electron. Lett.* **14** 194–5
- Makdissi A 2003 AlaVar 5.2 <http://www.alamath.com/index.html>
- Maruyama K, Mizushima S, Sugiura T, Van Leeuwen G M J, Hand J W, Marrocco G, Bardati F, Edwards A D, Azzopardi D and Land D 2000 Feasibility of noninvasive measurement of deep brain temperature in newborn infants by multifrequency microwave radiometry *IEEE Trans. Microw. Theor. Tech.* **48** 2141–7
- Meredith R, Warner F L, Davies Q V and Clark J L 1964 Superheterodyne radiometers for short millimetre wavelengths *Proc. IEE* **111** 241–56
- Mizushima S, Hamamamura Y, Matsuda M and Sugiura T 1989 A method of solution for a class of inverse problems involving measurement errors and its application to medical microwave radiometry *IEEE MTT-S Int. Microw. Symp. Digest* **1** 171–4
- Park B K, Kim J H and Kim K C 1991 Performance evaluation of an X-band RF switching radiometer *IEEE Trans. Instrum. Meas.* **40** 483–5
- Randa J 2001 Stability measurements on noise sources *IEEE Trans. Instrum. Meas.* **50** 368–72
- Roberts M J and Blalock T V 1985 Effect of filter frequency response in measuring statistical estimators of input noise *Rev. Sci. Instrum.* **56** 326–8
- Rutman J 1974 Characterization of frequency stability: a transfer function approach and its application to measurements via filtering of phase noise *IEEE Trans. Instrum. Meas.* **23** 40–8
- Schwartz M 1970 *Information Transmission, Modulation and Noise* (New York: McGraw-Hill) chapter 7
- Stelzried C T 1968 Microwave thermal noise standards *IEEE Trans. Microw. Theor. Tech.* **16** 646–55
- Sugiura T, Kouno Y, Hashizume A, Hirata H, Hand J W, Okita Y and Mizushima S 2004 Five-band microwave radiometer system for non-invasive measurement of brain temperature in new-born infants: system calibration and its feasibility *IEEE EMBS Proc. 26th Ann. Int. Conf.* pp 2292–5
- Terman F E 1955 *Electronic and Radio Engineering* (New York: McGraw-Hill) chapter 9
- Van der Ziel A 1955 *Noise* (London: Chapman and Hall) chapter 13
- Van der Ziel A 1976 *Noise in Measurements* (New York: Wiley-Interscience) chapter 7
- Wait D F 1967 The sensitivity of the Dicke radiometer *J. Res. Nat. Bur. Stand.* **71C** 127–52
- Wait D F 1968 Thermal noise from a passive linear multiport *IEEE Trans. Microw. Theor. Tech.* **16** 687–91
- Wait D F and Nemoto T 1968 Noise temperature of a mismatched noise source *IEEE Trans. Microw. Theor. Tech.* **16** 670–5
- Wells J S, Daywitt W C and Miller C K S 1964 Measurement of effective noise temperatures of microwave noise sources *IEEE Trans. Instrum. Meas.* **13** 17–28
- White R D 1989 The noise bandwidth of sampled data systems *IEEE Trans. Instrum. Meas.* **38** 1036–43
- Wiley R G 1977 A direct time-domain measure of frequency stability: the modified Allan variance *IEEE Trans. Instrum. Meas.* **26** 38–41